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The final prototype of the handlebar extension has been thoroughly developed and is explained with sufficient detail to ensure that objective data on nearly all design requirements can be measured or verified. This iteration reflects a comprehensive approach to addressing the functional and extra-functional criteria outlined in the project. The prototype incorporates multiple subsystems that have been carefully designed to meet the client's needs and the relevant engineering principles.

Design Iteration Details:

1. Materials Selection:

PVC Pipes: Used for the structural framework of the handle. The decision to use PVC was based on its flexibility, lightweight properties, and durability. This material allows for some give, which is important for comfort, but maintains enough rigidity to support the handle and withstand the pressure exerted when pushing the cart.

Wood: Chosen for the handle's main body due to its strength and aesthetic qualities. The wood is treated to prevent wear and ensure that it remains durable over time. The edges are rounded to meet ergonomic requirements and avoid discomfort when gripped.

Plastic Tubing for Grip: As per expert advice, the plastic tubing provides additional grip to the original handle of the cart and improves the ergonomics of the handle. With this change, a noticeable change in how much the extension shifted during use. This addition ensures that the handle is not only functional but also user-friendly for extended periods of use.

2. **Adjustability Mechanism:**

The handle is designed with an adjustable length feature that allows Mr. Kiser to modify the height of the handle based on his needs and the load being carried. This is achieved using an easily adjustable mechanism with hooks or sliding pipes, allowing the handle to extend or shorten without complicated tools or processes. This adjustability ensures that the handle can adapt to varying user heights and loads, addressing the client's specific requirement for a versatile solution.

3. **Attachment System:**

Clamps: In place of duct-taped Velcro, which was deemed insufficient during testing, the team utilized clamps to attach the handle to the cart securely, while allowing the velcro to be adjusted based on user-preference. This system ensures that the handle remains stable during use but can also be easily removed or adjusted when necessary. The attachment system has been modeled for strength and stability, ensuring that it can withstand the forces exerted during normal operation.

Wedges: Due to the free-falling flaw noticed during classmate tests, our team used the recommendations of stakeholder Mr. Kiser to add a wooden wedge to both posts. This addition allowed the handle to stay stationary without intervention from an outside source. With the ability to withstand gravity, this design's attachment system allows for comfortable use.



Attributes That Can Be Tested:

1. Ergonomics and Comfort:

Classmate Testing and Population Surveys: The design of the handle was tested for comfort using a hand-fitting prototype, and adjustments were made to ensure it is ergonomic. While mathematical modeling for ergonomics is more subjective, data from testing and feedback from users were gathered to validate that the design minimizes strain and discomfort. Comfort tests have been conducted, and it was confirmed that the handle does not cause discomfort during typical use.

2. Adjustability and Range of Motion:

Modeling of Adjustability: The adjustable mechanism was tested to ensure that the handle can extend and retract smoothly without compromising stability. While the exact range of motion can be modeled mathematically, physical testing of the mechanism ensures that it meets the height range needed for a variety of users and cart loads, while still exhibiting a wide range of usable angles.

Attributes Requiring Expert Review:

While many aspects of the design can be tested or modeled mathematically, some elements require expert review, as they cannot easily be quantified through traditional testing. For example:

Aesthetics and Design Appeal: The visual appeal of the handlebar extension is subjective and cannot be easily modeled. The design's aesthetic features were reviewed by experts in product design, including Mrs. Piper and Mr. Jared, whose feedback ensured that the handle met the client's requirements for a pleasing and polished appearance.

Long-Term Durability: While material selection (PVC, wood, and plastic tubing) ensures durability in the short term, long-term performance, such as wear and tear over five years, cannot be precisely modeled. However, experts in material science and product durability provided input on the suitability of the materials for prolonged use. Their reviews helped confirm that the chosen materials were appropriate for long-term use, with considerations for stress, environmental conditions, and wear resistance.

The final prototype iteration has been developed with sufficient detail and rigor to allow for objective testing and validation of most design requirements. Strength, load-bearing capacity,

ergonomics, and adjustability have all been modeled or tested to ensure functionality. For attributes that could not be modeled or tested through traditional methods, expert review provided the necessary feedback and confirmation to address potential pitfalls. This comprehensive approach ensures that the handlebar extension is both practical and viable for Mr. Kiser's needs, meeting safety, comfort, and durability standards while incorporating expert guidance to fine-tune the design.